Lei Zhang, Ph.D. student

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Research Interests

Gravitational lensing, AGN feedback, Galaxy evolution, Rapid quenching, High-z galaxies

Education

† *Indicates expected*

2022–Sep, 2026 †	Ph.D., School of Astronomy and Space Science, Nanjing University
	Supervisor: Prof. Zhi-Yu Zhang
2020–2022	M.Sc., School of Astronomy and Space Science, Nanjing University
	Supervisor: Prof. Zhi-Yu Zhang
2016–2020	B.Sc., School of Astronomy and Space Science, Nanjing University

Publications

[1] **L. Zhang**, Z.-Y. Zhang, J. W. Nightingale, Z.-C. Zou, X. Cao, C.-W. Tsai, C. Yang, Y. Shi, J. Wang, D. Xu, L.-R. Lin, J. Zhou, and R. Li, "Discovery of a radio jet in the Cloverleaf quasar at z = 2.56," *Mon. Not. R. Astron. Soc.*, vol. 524, no. 3, pp. 3671–3682, Sep. 2023. DOI: 10.1093/mnras/stad2069. arXiv: 2212.07027 [astro-ph.GA].

[2] **L. Zhang**, Z.-Y. Zhang, L.-R. Lin, A. Man, C. Yang, X. Cao, C.-W. Tsai, T. Wang, Z.-C. Zou, Y. Zhao, Y. Qiu, Y. Shi, J. Wang, and D. Xu, "Localized jet-suppressed star formation in a radio-quiet quasar", manuscript submitted.

[3] **L. Zhang**, Z.-Y. Zhang, L.-R. Lin, Y. Yang, A. Man, C.-W. Tsai, C. Yang, X. Cao, Z.-C. Zou, D. Xu, "A magnified view of major merger in a distant quasar", in preparation.

[4] K. Kade, C. Yang, M. Yttergren, K. K. Knudsen, S. König, A. Amvrosiadis, S. Dye, J. Nightingale, **L. Zhang**, Z. Zhang, A. Cooray, P. Cox, R. Gavazzi, E. Ibar, J. M. Michałowski, P. van der Werf, and R. Xue, "Detailed lens modeling and kinematics of the submillimeter galaxy G09v1.97. An analysis of CO, H2O, H2O+, and dust continuum emission", *Astron. Astrophy*, Jan. 2026. DOI: 10.48550/arXiv.2601.14685. arXiv: 2601.14685 [astro-ph.GA].

Telescope Time Awarded

PI of observing proposal	VLA / 24B-154 (13.3 hrs), VLA / 24A-212 (9.0 hrs), VLBA / BZ105 (16.0 hrs)

Skills

Astronomical Expertise

Radio/Sub-mm Astronomy	Observation Proposal (VLA, VLBA, ALMA, JWST, VLT/ERIS)
	Observation Scheduling (VLA & VLBA)
	Data Reduction (radio/sub-mm interferometer)
Lensing Model	Galaxy Lensing (proficient), Cluster Lensing (basic)
Radiative Transfer Model	myRadex (basic)
Astronomical Tools	PyAutoLens, CASA, Astropy, DS9, AIPS, LensTool, ^{3D}Barolo, Carta

Core Skills

Learn independently and apply knowledge in practice

High comfort level for meaningful or interesting things

Technical Skills

AI-related	Vibe coding, Agents, Sub-agents
Coding	Python (proficient), C++ (basic)
Computer-related	CLI, Docker, Network security, Reverse tunnelling, CI/CD, Web design, Frontend, Backend, Virtual machine, Virtual network mesh

Open Source Projects

Build Your Own Games!	Introduce vibe-coding game-develop to public (CI/CD, frontend, backend)

Awards

2016–2019	Top Talent Program Scholarship $\times 3$
2020–2022	Master's Academic Scholarship $\times 2$
2022–2026	PhD Academic Scholarship $\times 4$